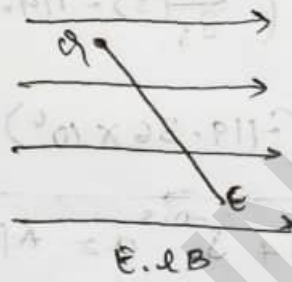
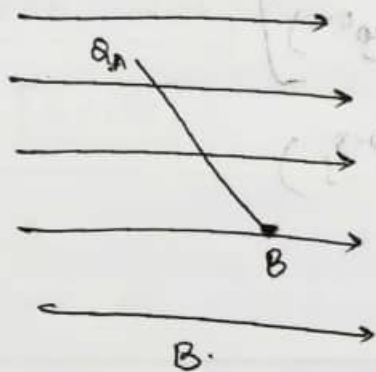


06/01/2023

MODULE-04.• FORCE ON MOVING POINT CHARGE / LORENTZ FORCE

A positive point charge moving with a velocity V in a uniform magnetic field of flux density B experiences a force which is given by, $F_1 = Q(V \times B) \longrightarrow (1)$.



If the point charge q is placed under the influence of electric field E the force on the charge is given by,

$$F_2 = QE \longrightarrow (2)$$

If the point charge q is placed under the influence of electric fields then the resultant force on q is

$$F = F_1 + F_2 \longrightarrow (3)$$

$$F = Q[V \times B + E]$$

• PROBLEM:

1. A point charge $Q = 18 \text{ nC}$ as a velocity of $V = 5 \times 10^6 \text{ m/sec}$ in the direction $\hat{A}_V = 0.6\hat{a}_x + 0.7\hat{a}_y + 0.3\hat{a}_z$.

Calculate the magnitude of force exerted on the charge under the influence of following fields

$$E = -3ax + 4ay + 6az \text{ kV/m}$$

$$B = -3ax + 4ay + 6az \text{ kV/m}$$

ii) Under the influence of Both fields.

Ans i) $F = QE$

$$= 18 \times 10^{-9} (-3ax + 4ay + 6az) \times 10^3$$

$$F = -54na_x + 72na_y + 108na_z$$

$$|F| = \sqrt{(-54 \times 10^{-9})^2 + (72 \times 10^{-9})^2 + (108 \times 10^{-9})^2}$$

$$= 140.58 \times 10^{-9} \times 10^3$$

$$|F| = 140.58 \times 10^{-6} \text{ N}$$

ii) $F = Q(V \times B)$

$$= Q(V_{av} \times B)$$

$$F = 18 \times 10^{-9} \times 5 \times 10^6 \begin{bmatrix} a_x & a_y & a_z \\ 0.6 & 0.95 & 0.3 \\ -3 & 4 & 6 \end{bmatrix} \text{ m/sec}$$

$$= 18 \times 10^{-3} \times 5 [a_x (4.5 - 1.2) - a_y (3.8 + 0.9) + a_z (2.24 + 1.25)] \times 10^{-3}$$

$$= 90 \times 10^{-3} [3.3a_x - 4.5 + 0.15a_z] \times 10^{-3}$$

$$= 90 \times 10^{-3} \sqrt{(3.3)^2 + (4.5)^2 + (1.65)^2} \times 10^{-3}$$

$$F = 653.94 \text{ N}$$

iii) $F = F_1 + F_2$

$$= 140.58 \times 10^{-6} + 653.94$$

$$F = 653.94 \text{ N}$$

~~Obtain an expression for Magnetic force on differential element (Q)~~
~~Derive an expression~~

- Obtain an expression for magnetic force on differential element (or)

Derive an expression for force on differential current carrying element (or)

Find the expression for force on differential current element moving in steady magnetic field.

→ The force on the point charge when placed in the magnetic field with flux density given by,

$$F = Q(\mathbf{v} \times \mathbf{B})$$

∴ The differential force on differential current element is given by,

$$dF = dQ(\mathbf{v} \times \mathbf{B}) \longrightarrow (1)$$

From the concept of current density

$$\mathbf{J} = S_v \mathbf{v} \text{ A/m}^2 \longrightarrow (2)$$

Since the differential current element dQ is given as $dQ = S_v dv \longrightarrow (3)$

Substitute (3) in (1)

$$dF = S_v dv (\mathbf{v} \times \mathbf{B}) \longrightarrow (4)$$

The eqn (4) can be re-written as

$$dF = S_v \mathbf{v} \times \mathbf{B} dv$$

from expression (2)

$$dF = \mathbf{J} \times \mathbf{B} dv \longrightarrow (5)$$

The relation between the differential current element in terms of surface current is

$$\mathbf{J} \cdot dv = I \cdot dl$$

The expression (5), can be re-written as

$$dF = I dl \times \mathbf{B} \longrightarrow (6)$$

The total force is given as

$$\int dF = \int I \cdot dl \times \mathbf{B}$$

$$\therefore F = I \int dl \times B$$

$$\boxed{F = IL \times B}$$

$$\boxed{F = (IL)(B \sin \theta \text{ in}) N} \text{ in vector form}$$

• Problems:

1. A conductor 4m long lies along y-axis with a current 10 amp in ay direction. Find the force on the conductor if the field in the region is $B = 0.005 a_x$ Tesla.

Ans $F = Idl \times B$

$$= (10)(4) a_y \times (0.005) a_x$$

$$= (0.2) (a_y \times a_x)$$

$$\boxed{F = -0.2 a_z N}$$

$$a_y \times a_x = -a_z$$

$$a_y \times a_y = 0$$

$$a_y \times a_z = a_x$$

2. The field $B = -2a_x + 3a_y + 4a_z$ T is present in free space. Find the vector force exerting in free space.



Ans $F = Idl \times B$

$$Idl = ?$$

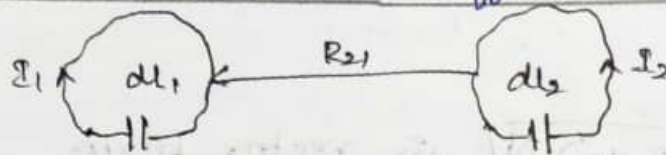
$$dl = dx$$

$$F = I dx \times (-2a_x + 3a_y + 4a_z)$$

$$\boxed{F = -2I + 3I a_z + 4I a_y}$$

$$F = -2I + 3I a_z + 4I a_y$$

Force Between two differential current elements



Consider 2 current loops with currents I_1 and I_2 . The loops are divided into small vector line segments dl and each current element is represented as $I \cdot dl$. Both current elements $I_1 dl_1$ and $I_2 dl_2$ produce the magnetic field.

The magnetic field produced by $I_1 dl_1$ and $I_2 dl_2$ is given by

$$dB_2 = \frac{\mu_0}{4\pi} \frac{I_2 dl_2}{R_{21}^2} \hat{a}_{21} \quad (1)$$

The force on the current element $I_1 dl_1$ and $I_2 dl_2$ due to dB_2 is given by,

$$\begin{aligned} dF &= I_1 dl_1 \times dB_2 \\ &= I_1 dl_1 \times \frac{\mu_0 I_2 dl_2}{4\pi R_{21}^2} \hat{a}_{21} \end{aligned}$$

$$dF_1 = \frac{\mu_0 I_1 dl_1 \times [I_2 dl_2 \times \hat{a}_{21}]}{4\pi R_{21}^2} \text{ N}$$

Similarly,

Force on current element $I_2 dl_2$ due to dB_1 ,

$$dF_2 = I_2 dl_2 \times dB_1$$

$$\text{where } dB_1 = \frac{\mu_0}{4\pi} \frac{I_1 dl_1}{R_{12}^2} \hat{a}_{12}$$

$$dF_2 = I_2 dl_2 \times \frac{\mu_0 I_1 dl_1}{4\pi R_{12}^2} \hat{a}_{12}$$

$$dF_2 = -\frac{\mu_0 I_2 dl_2 \times (I_1 dl_1 \times \hat{a}_{12})}{4\pi R_{12}^2} \text{ N}$$

$$F_1 = \frac{\mu_0 I_1 I_2}{4\pi} \int_{L_1} \int_{L_2} \frac{dl_1 \times (dl_2 \times a_{12})}{R_{12}^2} N$$

Similarly,

$$F_2 = \frac{\mu_0 I_1 I_2}{4\pi} \int_{L_1} \int_{L_2} \frac{dl_1 \times (dl_2 \times a_{12})}{R_{12}^2} N$$

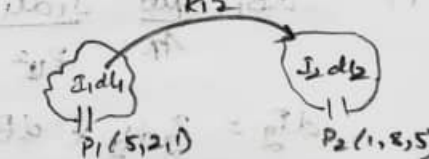
$$\therefore \boxed{F_1 = -F_2}$$

• PROBLEMS:

1. A current element $I_1 dl_1 = -3ay \text{ Am}$ at $P_1(5, 2, 1)$ and $I_2 dl_2 = -4az \text{ Am}$ at $P_2(1, 8, 5)$. Find differential force on dl_2 .

Ans $dB_1 = \frac{\mu_0}{4\pi} \frac{I_1 dl_1}{R_{12}^2} a_{12} \rightarrow (1)$

$dF_2 = I_2 dl_2 \times dB_1 \rightarrow (2)$



$$R_{12} = (1-5)ax + (8-2)ay + (5-1)az$$

$$R_{12} = -4ax + 6ay + 4az$$

$$|R_{12}| = \sqrt{(-1)^2 + (6)^2 + (4)^2} = \sqrt{16 + 36 + 16} = 8.24$$

$$a_{12} = \frac{R_{12}}{|R_{12}|} = \frac{-4ax + 6ay + 4az}{8.24}$$

$$dB_1 = \frac{\mu_0}{4\pi} \frac{(-3ay)}{(8.24)^2} \times (-4ax + 6ay + 4az)$$

$$dF_2 = -4az \times \left[\frac{\mu_0}{4\pi} \frac{(-3ay)}{(8.24)^2} \times (-4ax + 6ay + 4az) \right]$$

$$dF_2 = \frac{4\pi \times 10^{-7}}{4\pi} \frac{(4az) \times (-3ay) \times (-4az + 6ay + 4az)}{(8.24)^3}$$

$$= +12az \times (-4az + 6ay + 4az)$$

$$= 0 + 36az + 48az$$

$$dF_2 = 8.56 ay \text{ uN}$$

~~X = 36am~~

2. A Current element $I_1 dl_1 = 10^{-4} az$ Am located at $P_1(2, 0, 0)$. Another Current element $I_2 dl_2 = 10^{-6}(ax - 2ay + 3az)$ is located at $P_2(-2, 0, 0)$ and both of them are in free space

i) find force exerted on $I_2 dl_2$ by $I_1 dl_1$

ii) find force exerted on $I_1 dl_1$ by $I_2 dl_2$.

Ans

$I_1 dl_1 = 10^{-4} az$ at $P_1(2, 0, 0)$

$I_2 dl_2 = 10^{-6}(ax - 2ay + 3az)$ $P_2(-2, 0, 0)$

$$1) \quad dB_1 = \frac{\mu_0}{4\pi} \frac{I_1 dl_1}{R_{12}^2} a_{12}$$

$$dF_2 = I_2 dl_2 \times dB_1$$

$$R_{12} = -4ax$$

$$|R_{12}| = \sqrt{16} = 4$$

$$a_{12} = \frac{R_{12}}{|R_{12}|} = \frac{-4ax}{4}$$

$$dB_1 = \frac{\mu_0}{4\pi} \times \frac{10^{-4} az}{(4)^2} (-4ax)$$

$$= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{10^{-4} az \times (-4ax)}{4^3}$$

$$dB_1 = -0.625 \times 10^{-12} ay$$

$$dF_2 = I_2 dl_2 \times (-0.625 \times 10^{-12}).$$

$$= 10^{-6}(ax - 2ay + 3az) \times (-0.625 \times 10^{-12}(ay))$$

$$= -6.25 \times 10^{-19} az + 0 + 1.875 \times 10^{-18} ax$$

$$\boxed{dB_2 = -6.25 \times 10^{-19} az + 1.875 \times 10^{-18} ax}$$

$$ii) dB_2 = \frac{\mu_0}{4\pi} \frac{I_2 dl_2}{R_{22}^2} a_{22}$$

$$dF_2 = I_2 dl_2 \times dB_2$$

$$R_{22} = (2+2)ax.$$

$$R_{22} = 4ax.$$

$$|R_{22}| = \sqrt{4^2} = \frac{4}{1}.$$

$$a_{22} = \frac{R_{22}}{|R_{22}|} = \frac{4ax}{4}$$

$$dB_2 = \frac{\mu_0}{4\pi} \times \frac{10^{-6}(ax - 2ay + 3az) \times (4ax)}{(4)^3}$$

$$= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{10^{-6}ax - 2 \times 10^{-6}ay + 3 \times 10^{-6}az}{4^3} \times 4ax.$$

$$= 0 - 8 \times 10^{-13} az + 1.2 \times 10^{-12} ay$$

$$\boxed{dB_2 = 1.875 \times 10^{-11} ay - 1.25 \times 10^{-11} az}$$

$$dF_1 = I_1 dl_1 \times dB_2$$

$$= 10^{-4} az \times (1.875 \times 10^{-11} ay - 1.25 \times 10^{-11} az)$$

$$= 1.875 \times 10^{-8} ax - 1.25 \times 10^{-8}(0)$$

$$\boxed{dF_1 = -1.875 \times 10^{-8} ax}$$

Module 4(b):

Magnetic materials:

1. Magnetic pole strength
2. Magnetic moment (m)
3. Magnetization (M)
4. Susceptibility (χ)

1. Magnetic pole strength

The magnetic pole strength of a pole is said to be unity if it experiences a force of 1 Newton when it is separated from 1mtr from another pole.

2. Magnetic Moment (m).

It is the product of the magnetic pole strength and the distance between the two poles.

3. Magnetization (M).

Magnetic moment per unit volume of magnet is called magnetization.

$$\therefore M = \frac{m}{V}$$

4. Susceptibility (χ).

It is the ratio of magnetization to the magnetic field.

$$\therefore \chi = \frac{M}{H}$$

6. Permeability (μ):

$$\mu = \mu_0 \mu_r$$

$\mu_0 \rightarrow$ Absolute permeability. $= 4\pi \times 10^{-7} \text{ A/m}$

$\mu_r \rightarrow$ relative permeability $= 1$

* The relation Between B, H and M

The relation Between B and H,

$$B = \mu_0 H \longrightarrow (1)$$

The magnetic flux density, B due to extra flux in the medium is given by,

$$B = \mu_0 M \longrightarrow (2)$$

The resultant ~~is~~ magnetic flux density,

$$B = \mu_0 H + \mu_0 M$$

$$B = \mu_0 (H + M) \longrightarrow (3)$$

Since,

$$B = \mu H = \mu_0 \mu_r H$$

Substituting in eq (3) for B,

$$\mu_0 \mu_r H = \mu_0 (H + M)$$

$$\mu_r H = H + M$$

$$M = \mu_r H - H$$

$$M = H(\mu_r - 1) \longrightarrow (4)$$

Magnetization in terms of susceptibility,

$$\chi = \frac{M}{H}$$

$$M = \chi H$$

Substitute for M from (4)

$$\chi H = H(\mu_r - 1)$$

$$\chi = \mu_r - 1 \longrightarrow (5)$$

Problems:

1. Find the magnetic field intensity, inside a magnetic material for the following conditions

i) $M = 100 \text{ A/m}$, $\mu = 1.5 \times 10^{-5} \text{ H/m}$

ii) $B = 200 \mu\text{T}$, $\chi_m = 15$

Ans 1) $\chi = \frac{M}{H}$, $H = \frac{M}{\chi}$

$$\chi = \mu_r - 1$$

$$\mu = \mu_0 \mu_r$$

$$\mu_0 = 1.5 \times 10^{-5} \text{ H/m}$$

$$\mu_r = \frac{\mu}{\mu_0}$$

$$= \frac{1.5 \times 10^{-5}}{4\pi \times 10^{-7}}$$

$$\boxed{\mu_r = 118.936}$$

$$\chi = \mu_r - 1$$

$$= 118.936 - 1$$

$$\boxed{\chi = 10.893}$$

$$H = \frac{M}{\chi}$$

$$= \frac{100}{10.893} = \frac{100}{10.93}$$

$$\boxed{H = 9.14 \text{ A/m}}$$

ii) $B = 200 \mu\text{T}$, $\chi_m = 15$

$$\chi = \mu_r - 1$$

$$15 = \mu_r - 1$$

$$\boxed{16 = \mu_r}$$

$$B = \mu_0 \mu H$$

$$200 \mu = 4\pi \times 10^{-7} \times 16 \times H$$

$$H = 9.947 \text{ Am}$$

$$\frac{B}{\mu_0} = H \quad (1)$$

$$200 \mu$$

$$\frac{200 \times 10^{-6}}{4\pi \times 10^{-7}} =$$

$$159.15$$

$$159.15 \times 10^{-6} = H$$

2. Find the magnetization in a material where

i) $\mu = 1.8 \times 10^{-5} \text{ H/m}$ and $H = 120 \text{ A/m}$.

ii) $\mu_r = 22$ and the no. of atoms are $8.3 \times 10^{28} \text{ A/m}^3$ and each atom has electric dipole moment of $4.5 \times 10^{-29} \text{ A/m}^2$

iii) $B = 300 \mu\text{T}$ and $\chi = 15$.

Ans. i) $M = ?$ $M = \frac{M}{V}$

$$\mu = \mu_0 \mu_r H$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{1.8 \times 10^{-5}}{4\pi \times 10^{-7}} = 14.32$$

$$M = \chi H$$

$$M = (\mu_r - 1) H$$

$$M = (14.32 - 1)(120)$$

$$M = 1598.4 \text{ A/m}$$

ii) $\mu_r = 22$.

$$M = n \times m$$

$$n = 8.3 \times 10^{28}$$

$$m = 4.5 \times 10^{-29}$$

$$\therefore M = 8.3 \times 10^{28} \times 4.5 \times 10^{-29}$$

$$M = 373.5 \text{ A/m}$$

$$\begin{aligned} \text{ii) } \mu &= \frac{B}{\mu_0} \\ &= \frac{300 \times 10^{-6}}{4\pi \times 10^{-7}} \end{aligned}$$

$$\boxed{\mu = 238.732 \text{ A/m}}$$

2

P

W

u

i

ii

iii

iv

v

vi

vii

viii

ix

x

xi

xii

xiii

xiv

xv

xvi

xvii

xviii

xix

xx

3) If the magnetic flux $B = 0.05 \text{ x ay}$ Tesla in a material for which $\mu_r = 2.5$. Find

- i) μ_r ii) μ iii) H iv) M v) J vi) J_d

Ans i) $\mu_r = 2.5$

$$\begin{aligned} \mu_r &= 2.5 \\ &= 2.5 + 1 \end{aligned}$$

$$\boxed{\mu_r = 3.5}$$

ii) $\mu = \mu_0 \mu_r$

$$= 4\pi \times 10^{-7} \times 3.5$$

$$\boxed{\mu = 4.39 \times 10^{-6} \text{ H/m}}$$

iii) $H =$

$$B = \mu H$$

$$B = \mu_0 \mu_r H$$

$$H = \frac{B}{\mu_0 \mu_r}$$

$$H = \frac{0.05 \text{ x ay}}{4.39 \times 10^{-6}}$$

$$\boxed{H = 11.38 \times 10^3 \text{ x ay A/m}}$$

$$iv) M = M = 2mH$$

$$M = 2.5 \times 11.38 \times 10^3 \times a_y$$

$$M = 28.45 \times 10^3 \times a_y \text{ A/m}$$

$$v) J = \nabla \times H$$

$$= \begin{bmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{bmatrix}$$

$$= \begin{bmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 11.368 \times 10^3 & 0 \end{bmatrix}$$

$$= \frac{\partial}{\partial x} [11.368 \times 10^3 x] a_z$$

$$J = 11.368 \times 10^3 a_z \text{ A/m}^2$$

$$vi) J_d = \nabla \times M$$

$$= \begin{bmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 28.42 \times 10^3 & 0 \end{bmatrix}$$

$$= a_z \left(\frac{\partial}{\partial y} (28.42 \times 10^3 y) \right)$$

$$J_d = 28.42 \times 10^3 a_z \text{ A/m}^2$$

4. Through a suitable experiment on a magnetic material, the magnetic flux density B is found to be 1.2 Tesla. when $H = 300 \text{ A/m}$. when H is increased to 1500 A/m . the B field increased to 1.5 Tesla. what is the percentage change in the magnetization vector.

Ans

$$B = \mu H$$

$$B = \mu_0 \mu_r H$$

$$\mu_r = \frac{B}{\mu_0 H}$$

$$= \frac{1.2}{4\pi \times 10^{-7} \times 300}$$

$$\boxed{\mu_r = 3.18 \times 10^3}$$

$$\mu_r = \frac{B}{\mu_0 H_2}$$

$$= \frac{1.5}{4\pi \times 10^{-7} \times 1500}$$

$$\boxed{\mu_r = 795.77}$$

$$M = \chi_m H$$

$$= (\mu_r - 1) H$$

$$M = (\mu_r - 1) H_1 = (3.18 \times 10^3 - 1) (300)$$

$$\underline{M_1 = 953.7 \times 10^3}$$

$$M = (\mu_r - 1) H_2 = (795.77 - 1) (1500)$$

$$\underline{M_2 = 1.19 \times 10^6}$$

$$\% \text{ge} = \frac{M_2 - M_1}{M_1} = \frac{1.19 \times 10^6 - 953.7 \times 10^3}{953.7 \times 10^3} \times 100$$

$$\boxed{\% \text{ge} = 24.77\%}$$

The Magnetic Boundary Conditions.

They are the conditions that the magnetic field intensity H and the magnetic flux density B should be such that

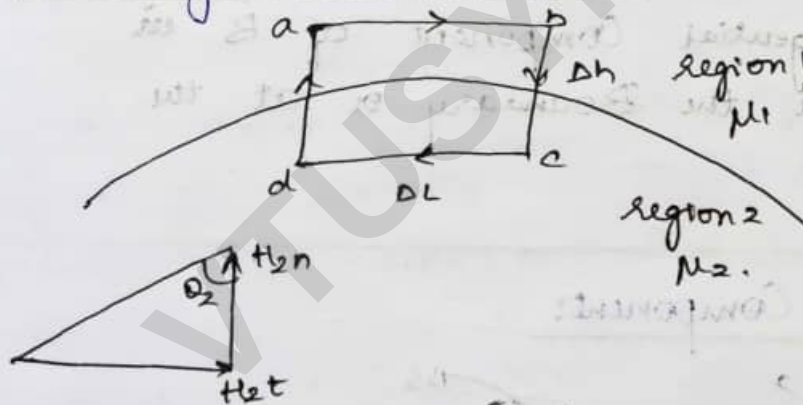
$$\oint_S B \cdot ds = 0$$

$$\oint H \cdot dl = I \text{ amps}$$

For the magnetic field, there are two components.

- (i) The tangential component
- (ii) The normal component.

1. The tangential Component:



Applying the ampere's circuital law, To the closed path ABCD. we can write

$$H_{1t} \cdot \Delta L - H_{2t} \cdot \Delta L = K \Delta L \longrightarrow (1)$$

Here $K \rightarrow$ the surface current density

$$\Delta L (H_{1t} - H_{2t}) = K \Delta L$$

$$K = H_{1t} - H_{2t} \longrightarrow (2)$$

Since $B = \mu H$, eqn (2) can be re-written as

$$k = \frac{B_{1t}}{\mu_1} - \frac{B_{2t}}{\mu_2} \longrightarrow (3)$$

The general form of expression (2) is

$$(H_1 - H_2) \times a_{n12} = k A/m$$

Here, the unit vector a_{n12} is normal to the interface and is directed from medium 1 to 2.

If the Boundary is free of current or media which is not a conductor, then $k=0$.

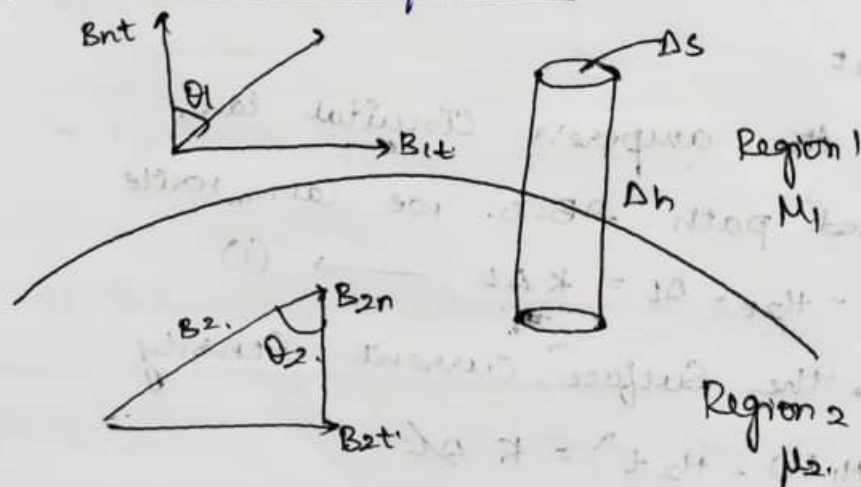
Therefore, the expression (3) becomes

$$\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2}$$

$$H_{1t} = H_{2t}$$

$\therefore$ The tangential component H is continuous while the tangential component of B is discontinuous at the boundary or at the interface.

2. The normal Component:



According to Gauss law for magnetic field.

$$\oint \mathbf{B} \cdot d\mathbf{s} = 0.$$

$$\therefore B_1 \Delta s - B_2 \Delta s = 0.$$

Therefore,

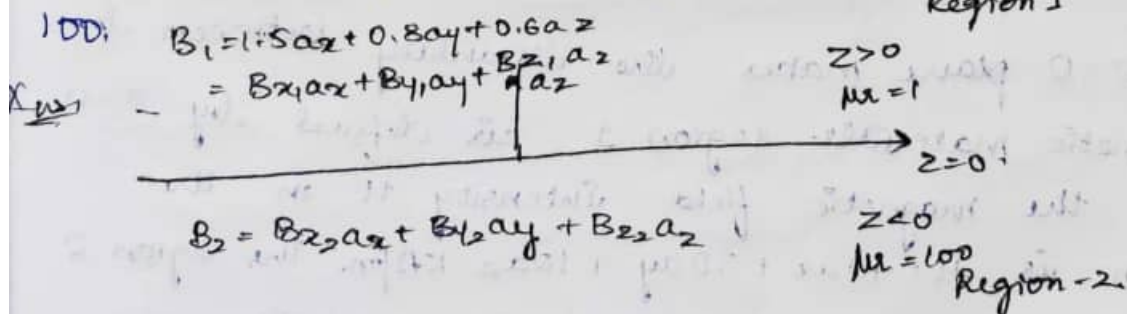
$$B_1 n = B_2 n$$

$$\mu_1 H_1 n = \mu_2 H_2 n.$$

This shows that, the normal component of $\mathbf{B}$ is continuous at the boundary whereas the normal component of $\mathbf{H}$ is discontinuous at the boundary.

PROBLEM!

1. The $z=0$ plane makes the boundary between the two magnetic material. Region 1 characterised by $z > 0$, region 2 characterised by $z < 0$. The magnetic flux density in region 1 is $\mathbf{B}_1 = 1.5\mathbf{a}_x + 0.8\mathbf{a}_y + 0.6\mathbf{a}_z$ mT. Find $\mathbf{B}$ for region 2. Assume the region 1 as free space and the relative permeability of region 2 as 100.



$$\mu n \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{K} \longrightarrow (1)$$

$$B_{n1} = B_{n2} \longrightarrow (2)$$

$$B_{z1} = B_{z2} \quad ; \quad B_{z1} = 0.6$$

$$B_{z2} = 0.6$$

Note:

$$\mathbf{B} = \mu \mathbf{H}$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu} = \frac{\mathbf{B}}{\mu_0 \mu_r}$$

$$a_z \times \left(\frac{B_1}{\mu_0 \mu_{r1}} - \frac{B_2}{\mu_0 \mu_{r2}} \right) = 0.$$

$$a_z \times \left(\frac{B_1}{\mu_0(1)} - \frac{B_2}{\mu_0 \times 100} \right) = 0$$

$$a_z \times \left(\frac{100 B_1 - B_2}{\mu_0 \times 100} \right) = 0.$$

$$a_z \times (100 B_1 - B_2) = 0.$$

$$a_z \times [100 (B_{x1} a_x + B_{y1} a_y + B_{z1} a_z) - (B_{x2} a_x + B_{y2} a_y + B_{z2} a_z)] = 0$$

$$a_z \times [100 B_{x1} - B_{x2} a_x + (100 B_{y1} - B_{y2}) a_y + (100 B_{z1} - B_{z2}) a_z] = 0.$$

$$100 B_{x1} - B_{x2} a_x + (100 B_{y1} - B_{y2}) a_y + (100 B_{z1} - B_{z2}) a_z = 0$$

$$(100(1.5) - B_{x2}) a_y + (100(0.8) - B_{y2}) a_x = 0.$$

$$(150 - B_{x2}) a_y + (80 - B_{y2}) a_x = 0.$$

$$150 - B_{x2} = 0$$

$$80 - B_{y2} = 0$$

$$B_{x2} = 150 \text{ mT}$$

$$B_{y2} = 80 \text{ mT}$$

$$B_{z2} = 0.6 \text{ mT}$$

$$B_2 = 150 a_x + 80 a_y + 0.6 a_z //$$

2. $z=0$ plane makes the boundary between two magnetic materials. region 1 is defined by $z > 0$ and the magnetic field intensity H in the region is $H = 40 a_x + 50 a_y + 10 a_z$ K A/m. The region 2 is defined by $z < 0$ and has relative permeability of 1000. if the relative permeability of medium is 200 find H in the medium 2

A current sheet of 12 ay kA/m is present at the boundary.

Ans

$$H_1 = 40ax + 50ay + 12az$$

Region 1

$$z > 0$$

$$\mu = 200$$

$$z < 0$$

$$\mu = 1000$$

Region 2

$$H_2 = ?$$

$$\hat{a}_n \times (H_1 - H_2) = K$$

$$B_{n1} = B_{n2}$$

$$B_{z1} = B_{z2}$$

$$H_{z1} \mu_{r1} = H_{z2} \mu_{r2}$$

$$H_{z2} = \frac{H_{z1} \mu_{r1}}{\mu_{r2}}$$

$$H_{z2} = \frac{(12)(200)}{1000}$$

$$H_{z2} = 2.4 \text{ kA/m}$$

$$\hat{a}_n \times (H_1 - H_2) = K$$

$$\hat{a}_z \times [(H_{x1} - H_{x2})\hat{a}_x + (H_{y1} - H_{y2})\hat{a}_y + (H_{z1} - H_{z2})\hat{a}_z] = 12\hat{a}_y$$

$$(H_{x1} - H_{x2})\hat{a}_y + (H_{y1} - H_{y2})\hat{a}_x + 0 = 12\hat{a}_y$$

$$(40 - H_{x2})\hat{a}_y + (50 - H_{y2})\hat{a}_x = 12\hat{a}_y$$

$$(40 - H_{x2})\hat{a}_y = 12\hat{a}_y$$

$$H_{x2} = 40 - 12$$

$$H_{x2} = 28 \text{ kA/m}$$

$$(50 - H_{y2})\hat{a}_x = 0$$

$$H_{y2} = 50 \text{ kA/m}$$

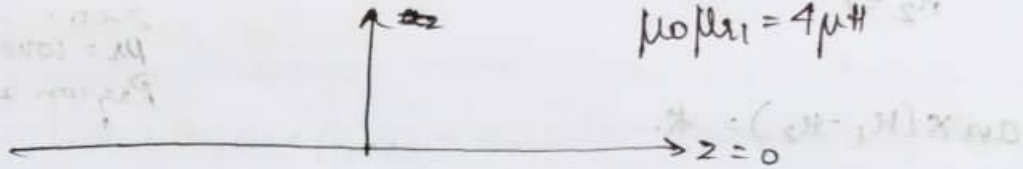
$$H_{z2} = 2.4 \text{ kA/m}$$

$$H_2 = 28ax + 50ay + 2.4az$$

3. For a region 1, $\mu_1 = 4\mu_0$ and $\mu_2 = 6\mu_0$ for region 2. The regions are separated by $z=0$ plane, the surface current density at the boundary is $K = 100 \text{ ax A/m}$. Find B_2 if $B_1 = 2a_x + 3a_y + a_z \text{ mT}$ for $z > 0$.

Ans

$$B_1 = 2a_x + 3a_y + a_z$$



$$B_2 = ?$$

$$\mu_0 \mu_{r2} = 6\mu_0$$

$$a_z \times \left(\frac{B_1}{\mu_0 \mu_{r1}} - \frac{B_2}{\mu_0 \mu_{r2}} \right) = 100 a_x$$

$$a_z \times \left(\frac{B_1}{4\mu_0} - \frac{B_2}{6\mu_0} \right) = 100 a_x$$

$$a_z \times \frac{3\mu_0 B_1 - 2\mu_0 B_2}{12\mu_0} = 100 a_x$$

$$a_z \times (3\mu_0 B_1 - 2\mu_0 B_2) = 1200\mu_0 a_x$$

$$a_z \times (B_1 - B_2)$$

$$a_z \times [3\mu_0 (B_{x1}a_x + B_{y1}a_y + B_{z1}a_z) - 2\mu_0 (B_{x2}a_x + B_{y2}a_y + B_{z2}a_z)] = 1200\mu_0 a_x$$

$$= a_z \times [(3\mu_0 B_{x1} - 2\mu_0 B_{x2})a_x + (3\mu_0 B_{y1} - 2\mu_0 B_{y2})a_y + (3\mu_0 B_{z1} - 2\mu_0 B_{z2})a_z] = 1200\mu_0 a_x$$

$$= a_z [(3\mu_0 B_{x1} - 2\mu_0 B_{x2})a_y + (3\mu_0 B_{y1} - 2\mu_0 B_{y2})a_x] = 1200\mu_0 a_x$$

$$(3\mu_0(2) - 2\mu_0 B_{x2})a_y + (3\mu_0(3) - 2\mu_0 B_{y2})a_x = 1200\mu_0 a_x$$

$$6\mu - 2\mu B_{x2} = 0$$

$$B_{x2} = \frac{6\mu}{2\mu}$$

$$B_{x2} = 3 \text{ A/m}$$

$$9\mu - 2\mu B_{y2} = 1200\mu$$

$$9\mu - 1200\mu = 2\mu B_{y2}$$

$$B_{y2} = \frac{9\mu - 1200\mu}{2\mu}$$

$$B_{y2} = -595.5 \text{ A/m}$$

$$B_2 = 3a_x - 595.5a_y + a_z \text{ mT}$$

POTENTIAL ENERGY AND FORCES ON MAGNETIC MATERIALS.

1) Derive the equation for energy density/potential energy in magnetic field.

a) Magnetic Energy:

The energy stored in the inductor is given by

$$W_m = \frac{1}{2} LI^2 \text{ J} \longrightarrow (1)$$

The expression for energy in electrostatics is given by,

$$W_E = \frac{1}{2} \int (\mathbf{D} \cdot \mathbf{E}) dv \text{ J} \longrightarrow (2)$$

Similarly,

Energy stored in magnetostatic,

$$W_m = \frac{1}{2} \int \mathbf{B} \cdot \mathbf{H} dv \longrightarrow (3)$$

But,

$$\mathbf{B} = \mu \mathbf{H}.$$

Substitute $\mathbf{B}$ in eq (3)

$$W_m = \frac{1}{2} \int \mu H^2 dv \longrightarrow (4).$$

Similarly,

The energy in magnetostatic can also be written as,

$$W_M = \frac{1}{2} \int_V \mu \cdot \frac{B^2}{\mu^2} dv.$$

$$W_M = \frac{1}{2} \int_V \frac{B^2}{\mu} dv. \longrightarrow (5)$$

* b) Magnetic Energy density.

It is the magnetic energy stored per unit volume.

$$e = \frac{WH}{V}$$

$$e = \frac{WB}{V}$$

$$e_m = \frac{1}{2} \mu H^2$$

$$e_m = \frac{1}{2} \frac{B^2}{\mu} \text{ J/m}^3$$

* c) Force on Magnetic materials.

$$F = \frac{B^2 S}{2\mu} N$$

S $\rightarrow$ Surface area

$\mu \rightarrow$ permeability.

* d) Magnetic force per unit area.

$$\frac{F}{S} = \frac{B^2}{2\mu} \text{ N/m}^2$$

$$\frac{F}{S} = \frac{\mu^2 H^2}{2\mu}$$

$$\frac{F}{S} = \frac{\mu H^2}{2}$$

→ Questions:

1. Find the magnetic field intensity within a magnetic material

i) where $M = 150 \text{ A/m}$ & $\mu = 1.5 \times 10^{-5} \text{ H/m}$.

ii) $B = 300 \mu\text{T}$ and $\chi = 15$

2. A current element $I \cdot dl = 10^{-4} \text{ A}$

Ans i) $\mu = \frac{M}{H} = \frac{1.5 \times 10^{-5}}{4\pi \times 10^{-7}} = 11.936$

$$\mu = \chi_m \mu_0 = (\mu_r - 1) \mu_0$$

$$H = \frac{M}{(\mu_r - 1)} = \frac{150}{(11.936 - 1)} = 13.2154 \text{ A/m}$$

$$\boxed{H = 13.2154 \text{ A/m}}$$

ii) the magnetic flux density is given by,

$$B = \mu H = \mu_0 \mu_r H \text{ Wb/m}^2$$

$$\mu_r = 1 + \chi_m$$

$$B = \mu_0 (1 + \chi_m) H \text{ Wb/m}^2$$

$$H = \frac{B}{\mu_0 (1 + \chi_m)} = \frac{300 \times 10^{-6}}{4\pi \times 10^{-7} (1 + 15)}$$

$$\boxed{H = 14.92 \text{ A/m}}$$